EE3110 - Probability Foundations for Electrical Engineers
Tutorial - 6

1. Let X_1 and X_2 be independent Poisson random variables with parameters $\lambda_1 > 0$ and $\lambda_2 > 0$, respectively. Define $X = X_1 + X_2$. Find the p.m.f. of X. Find the expected value and variance of X.

Solution:

- To find p.m.f of $X = X_1 + X_2$,

$$\begin{aligned}
p_X(k) &= \mathbb{P}[X = k] = \mathbb{P}[X_1 + X_2 = k] \\
&= \mathbb{P}\left(\bigcup_j \{X_1 = j, X_2 = k - j\}\right) \\
&= \sum_j \mathbb{P}[X_1 = j, X_2 = k - j] \quad \text{(by countable additivity)} \\
&= \sum_j \mathbb{P}[X_1 = j]\mathbb{P}[X_2 = k - j] \quad \text{(by independence)} \\
&= \sum_j \frac{e^{-\lambda_1} \cdot \lambda_1^j}{j!} \cdot \mathbb{1}_{\{j \geq 0\}} \cdot \frac{e^{-\lambda_2} \cdot \lambda_2^{(k-j)}}{(k-j)!} \cdot \mathbb{1}_{\{k-j \geq 0\}} \\
&= \sum_{j=0}^{k} e^{-(\lambda_1+\lambda_2)} \cdot \frac{\lambda_1^j \cdot \lambda_2^{k-j}}{j! \cdot (k-j)!} \cdot \frac{k!}{k!} \\
&= \frac{e^{-(\lambda_1+\lambda_2)}}{k!} \sum_{j=0}^{k} \binom{k}{j} \cdot (\lambda_1)^j \cdot (\lambda_2)^{(k-j)} \\
p_X(k) &= \frac{e^{-(\lambda_1+\lambda_2)} \cdot (\lambda_1 + \lambda_2)^k}{k!}
\end{aligned}$$

  Therefore, $X \sim \text{Poisson}(\lambda_1 + \lambda_2)$, the sum of two independent Poisson-distributed random variables, each with parameters λ_1 and λ_2, is itself a Poisson-distributed random variable with a parameter equal to the sum of the individual parameters.

- Expected value of $X = X_1 + X_2$, let $\lambda = \lambda_1 + \lambda_2$ then $X \sim \text{Poisson}(\lambda)$

$$\begin{aligned}
E[X] &= \sum_{k=0}^{\infty} k \cdot \frac{e^{-\lambda} \cdot \lambda^k}{k!} \\
&= \sum_{k=1}^{\infty} \frac{e^{-\lambda} \cdot \lambda^k}{(k-1)!} \\
&= \lambda \cdot \sum_{k=1}^{\infty} \cdot \frac{e^{-\lambda} \cdot \lambda^{k-1}}{(k-1)!} \\
&= \lambda \cdot 1 = \lambda_1 + \lambda_2
\end{aligned}$$

- Variance of X, $\text{Var}(X) = E[X^2] - (E[X])^2$

$$\begin{aligned} E[X^2] &= \sum_{k=0}^{\infty} k^2 \cdot \frac{e^{-\lambda} \cdot \lambda^k}{k!} \\ &= \lambda \sum_{k=1}^{\infty} k \cdot \frac{e^{-\lambda} \cdot \lambda^{k-1}}{(k-1)!} \\ &= \lambda \sum_{m=0}^{\infty} (m+1) \cdot \frac{e^{-\lambda} \cdot \lambda^m}{m!} \quad \text{where } m = k-1 \\ &= \lambda(\lambda+1) \\ \text{Var}(X) &= \lambda^2 + \lambda - \lambda^2 = \lambda = \lambda_1 + \lambda_2 \end{aligned}$$

2. Suppose that $X_1, X_2, \ldots$ are i.i.d. (independent and identically-distributed) random variables, each of which have the moment generating function $\psi(t)$. Let $Y = X_1 + \cdots + X_N$, where the number of terms N in this sum is a random variable having the Poisson distribution with mean λ. Assume that N and $X_1, X_2, \ldots$ are independent, and $Y = 0$ if $N = 0$. Determine the m.g.f. of Y.

Solution:

$$\begin{aligned} \dot{\psi}_Y(t) &= E(\exp(tY)) = E\left[\exp\left(tX_1 + \cdots + tX_N\right)\right] \\ &= E\left\{E\left[\exp\left(tX_1 + \cdots + tX_N\right) \mid N\right]\right\} \\ &= E\left\{[\psi(t)]^N\right\} \\ &= \sum_{x=0}^{\infty} [\psi(t)]^x \frac{\exp(-\lambda)\lambda^x}{x!} \\ &= \exp(-\lambda) \sum_{x=0}^{\infty} \frac{[\lambda\psi(t)]^x}{x!} \\ &= \exp(-\lambda)\exp(\lambda\psi(t)) = \exp\{\lambda[\psi(t)-1]\} \end{aligned}$$

3. Let N be a geometric random variable with parameter p and X_i, $i = 1, 2, \ldots$, be *i.i.d* Poisson random variables with mean λ. Find the p.m.f of $Y = \sum_{i=1}^{N} X_i$, i.e. Y is a random variable obtained by taking a sum of geometric number of independent Poisson random variables. You can leave the answer as a summation.

Solution:

$$\mathbb{P}(Y = y) = \sum_{n=1}^{\infty} \mathbb{P}(Y = y | N = n)\mathbb{P}(N = n)$$

The above is simply the Law of Total Probability. Now since N and all other X_i are independent, the conditioning reduces to the probability that $Y = y$ where now Y is simply the sum of exactly "n" independent Poissons.
Please note that sum of two independent Poisson random variable is again a Poisson with rate equal

to sum of rates.

$$\begin{aligned}\mathbb{P}(Y=y) &= \sum_{n=1}^{\infty} \mathbb{P}(X_1+X_2+X_3+.....X_n=y)\mathbb{P}(N=n) \\ &= \sum_{n=1}^{\infty} \frac{(n\lambda)^y e^{-n\lambda}}{y!}(1-p)^{n-1}p \\ &= \frac{\lambda^y p}{y!(1-p)} \sum_{n=1}^{\infty} e^{-n\lambda} n^y (1-p)^n \\ &= \frac{\lambda^y p}{y!(1-p)} \sum_{n=1}^{\infty} [e^{-\lambda}(1-p)]^n n^y\end{aligned}$$

Since $e^{-\lambda}(1-p)$ is less than 1 this sum converges for any value of y Its called the Poly-Logarithmic series and doesn't have a closed form.
The objective of this exercise was to deal with the idea of sum of a random number of numbers.

4. Heights (in inches) of women in a population are normally distributed with mean 65 and standard deviation 1. Heights (in inches) of men in the same population are normally distributed with mean 68 and standard deviation 3. One woman and one man are selected independently at random. Let W denote the woman's height and M the man's height. Find the probability that the woman is taller than the man, i.e. find $\Pr(W > M)$.

   (Use the standard normal random variable Z. The Z-score you will need is ($\Phi(0.94868) \approx 0.8289$)

Solution: Let $D := W - M$. Since W and M are independent normals,

$$D \sim N(\mu_D, \sigma_D^2), \qquad \mu_D = E[W] - E[M] = 65 - 68 = -3,$$

and

$$\sigma_D^2 = \mathrm{Var}(W) + \mathrm{Var}(M) = 1^2 + 3^2 = 10.$$

We want

$$\Pr(W > M) = \Pr(D > 0).$$

Standardize D using a standard normal Z:

$$\Pr(D > 0) = \Pr\left(\frac{D - \mu_D}{\sigma_D} > \frac{0 - (-3)}{\sqrt{10}}\right) = \Pr\left(Z > \frac{3}{\sqrt{10}}\right).$$

The required Z-score is $\dfrac{3}{\sqrt{10}} \approx 0.948683298$. Using the standard normal c.d.f. Φ,

$$\Pr\left(Z > \frac{3}{\sqrt{10}}\right) = 1 - \Phi\left(\frac{3}{\sqrt{10}}\right) \approx 1 - \Phi(0.94868).$$

From standard normal tables (or a calculator), $\Phi(0.94868) \approx 0.8289$. Hence

$$\Pr(W > M) \approx 1 - 0.8289 = 0.1711.$$

$$\boxed{\Pr(W > M) \approx 0.171}$$

5. Suppose $X_1, \ldots, X_n$ are i.i.d. Bernoulli(p) and let $S = \sum_{i=1}^{n} X_i$. For $k = 1, \ldots, n$, compute $\mathbb{P}(X_1 = 1 \mid S = k)$.

Solution: By independence,

$$\mathbb{P}(X_1 = 1 \mid S = k) = \frac{\mathbb{P}(X_1 = 1,\, S = k)}{\mathbb{P}(S = k)} = \frac{\mathbb{P}(X_1 = 1,\, X_2 + \cdots + X_n = k - 1)}{\mathbb{P}(S = k)}.$$

Now

$$\mathbb{P}(X_1 = 1,\, X_2 + \cdots + X_n = k - 1) = p \binom{n-1}{k-1} p^{k-1}(1-p)^{n-k},$$

and

$$\mathbb{P}(S = k) = \binom{n}{k} p^k (1-p)^{n-k}.$$

Hence

$$\mathbb{P}(X_1 = 1 \mid S = k) = \frac{p\binom{n-1}{k-1} p^{k-1}(1-p)^{n-k}}{\binom{n}{k} p^k (1-p)^{n-k}} = \frac{\binom{n-1}{k-1}}{\binom{n}{k}} = \frac{k}{n}.$$

Therefore,

$$\boxed{\mathbb{P}(X_1 = 1 \mid S = k) = \frac{k}{n} \quad (k = 1, \ldots, n).}$$

Interestingly, this is independent of p.

6. Let $X \sim \mathrm{Bin}(n, p)$ with probability mass function

$$f(x) = \mathbb{P}(X = x) = \binom{n}{x} p^x (1-p)^{n-x}, \qquad x = 0, 1, \ldots, n.$$

Find the mode(s) of X, i.e. the value of x for which $f(x)$ is maximum.

Solution: Consider the ratio

$$R_x = \frac{f(x+1)}{f(x)} = \frac{\binom{n}{x+1} p^{x+1}(1-p)^{n-x-1}}{\binom{n}{x} p^x (1-p)^{n-x}} = \frac{n-x}{x+1} \cdot \frac{p}{1-p}, \qquad x = 0, 1, \ldots, n-1.$$

Then:

$$\begin{cases} f(x+1) > f(x) & \iff R_x > 1, \\ f(x+1) = f(x) & \iff R_x = 1, \\ f(x+1) < f(x) & \iff R_x < 1. \end{cases}$$

Solve $R_x > 1$:

$$\frac{n-x}{x+1} \cdot \frac{p}{1-p} > 1 \iff (n-x)p > (x+1)(1-p) \iff x < (n+1)p - 1.$$

Similarly, $R_x < 1$ is equivalent to $x > (n+1)p - 1$, and $R_x = 1$ is equivalent to $x = (n+1)p - 1$. Hence $f(x)$ increases with x up to the largest integer strictly less than $(n+1)p - 1$, then decreases after the smallest integer strictly greater than $(n+1)p - 1$. Therefore,

$$\mathrm{mode}(X) = \begin{cases} \lfloor (n+1)p \rfloor, & \text{if } (n+1)p \notin \mathbb{Z}, \\ \{(n+1)p - 1,\ (n+1)p\}, & \text{if } (n+1)p \in \mathbb{Z}, \end{cases}$$

with edge cases $p = 0$ giving mode 0 and $p = 1$ giving mode n.

7. Let X be the number of distinct birthdays in a group of 110 people (i.e., the number of days in a year such that at least one person in the group has that birthday). Under the usual assumptions (no February 29, all the other 365 days of the year are equally likely, and the day when one person is born is independent of the days when the other people are born), find the mean and variance of X.

Solution: Let I_j be the indicator r.v. for the event that at least one of the people was born on the jth day of the year, so

$$X = \sum_{j=1}^{365} I_j \quad \text{with} \quad I_j \sim \text{Bern}(p), \quad \text{where} \quad p = 1 - \left(\frac{364}{365}\right)^{110}.$$

The I_j's are dependent but by linearity, we still have

$$E(X) = 365p \approx 95.083.$$

By symmetry, the variance is

$$\text{Var}(X) = 365\text{Var}(I_1) + 2\binom{365}{2}\text{Cov}(I_1, I_2).$$

To get the covariance, note that

$$\text{Cov}(I_1, I_2) = E(I_1 I_2) - E(I_1)E(I_2) = E(I_1 I_2) - p^2,$$

and

$$E(I_1 I_2) = P(I_1 I_2 = 1) = P(A_1 \cap A_2),$$

where A_j is the event that at least one person was born on the jth day of the year. The probability of the complement is

$$P(A_1^c \cup A_2^c) = P(A_1^c) + P(A_2^c) - P(A_1^c \cap A_2^c) = 2\left(\frac{364}{365}\right)^{110} - \left(\frac{363}{365}\right)^{110},$$

so

$$\text{Var}(X) = 365p(1-p) + 365 \cdot 364 \cdot \left(1 - 2\left(\frac{364}{365}\right)^{110} + \left(\frac{363}{365}\right)^{110} - p^2\right) \approx 10.019.$$

8. A total of 20 bags of Haribo gummy bears are randomly distributed to the 20 students in a certain Stat 110 section. Each bag is obtained by a random student, and the outcomes of who gets which bag are independent. Find the average number of bags of gummy bears that the first three students get in total, and find the average number of students who get at least one bag.

Solution:

Let X_j be the number of bags of gummy bears that the jth student gets, and let I_j be the indicator of $X_j \geq 1$. Then $X_j \sim \text{Bin}\left(20, \frac{1}{20}\right)$, so $E\left(X_j\right) = 1$. So $E\left(X_1 + X_2 + X_3\right) = 3$ by linearity. The average number of students who get at least one bag is

$$E\left(I_1 + \cdots + I_{20}\right) = 20E\left(I_1\right) = 20P\left(I_1 = 1\right) = 20\left(1 - \left(\frac{19}{20}\right)^{20}\right)$$

9. (a) Do there exist two discrete random variables X and Y such that X and Y have the same distribution, i.e, same PMF and CDF, but the event $X = Y$ never occurs? If yes, give an example. Assume same sample space.

   (b) Let's consider a simplified gacha mechanic, where you spend *tickets* to obtain two types of rewards: a Standard reward and a Premium reward. In one *pull*, you spend one *ticket* to receive a reward, which is either Standard or Premium with equal probability 1/2. The reward from a *pull* is independent of other *pulls*. You decide to keep *pulling* till you obtain at least one Standard and one Premium reward and then stop. What is the expected number of *pulls*? Assume that in each *pull* a reward is definitely obtained and that you have infinite *tickets*.

Solution:

(a) One such example is $X \sim \text{Bernoulli}(1/2)$. Set $Y = 1 - X$. Y is also Bernoulli(1/2) by symmetry, but X and Y are never equal.

(b) Denote by X the number of *tickets* needed, starting with the second *pull*, to obtain a reward different from what was received in the first *pull*. $X - 1$ is then Geo(1/2), so, $\mathbb{E}[X] = 2$. This is excluding the first *pull*. Including the first *pull*, expected total number of *pulls* is $\mathbb{E}[X + 1] = 3$.

10. **2D transformation of a bivariate Gaussian RV:** Let the joint PDF of two Gaussian random variables X, Y be given as

$$f_{XY}(x, y) = \frac{1}{2\pi(1-\rho^2)} \exp\left[-\left(\frac{x^2 - 2\rho xy + y^2}{2(1-\rho^2)}\right)\right].$$

Compute the joint PDF $f_{VW}(v, w)$, where

$$V = \frac{1}{2}(X^2 + Y^2),$$

$$W = \frac{1}{2}(X^2 - Y^2).$$

Solution: To find the PDF of the transformed RVs, we need to first find the Jacobian:

$$J = \begin{bmatrix} \frac{\partial V}{\partial X} & \frac{\partial V}{\partial Y} \\ \frac{\partial W}{\partial X} & \frac{\partial W}{\partial Y} \end{bmatrix} = \begin{bmatrix} X & Y \\ X & -Y \end{bmatrix}$$

Hence, the absolute determinant the Jacobian is given as:

$$|\det(J)| = 2|XY|$$

Also note that for a given (v, w), there are 4 ordered pairs $(x, y), (x, -y), (-x, y), (-x, -y)$ which give the same (v, w) after transformation. We have $x = \sqrt{v + w}, y = \sqrt{v - w}$. All these points give the same absolute determinant of the Jacobian too. Based on all these observations, using the PDF transformation formula, we have:

$$f_{VW}(v, w) = \frac{f_{XY}(x, y) + f_{XY}(-x, y) + f_{XY}(x, -y) + f_{XY}(-x, -y)}{2|xy|}$$

$$= \frac{1}{4\pi(1-\rho^2)\sqrt{v^2 - w^2}} \left(2\exp\left[-\left(\frac{2v - 2\rho\sqrt{v^2 - w^2}}{2(1-\rho^2)}\right)\right] 2\exp\left[-\left(\frac{2v + 2\rho\sqrt{v^2 - w^2}}{2(1-\rho^2)}\right)\right]\right)$$

$$= \frac{1}{2\pi(1-\rho^2)\sqrt{v^2-w^2}} \exp\left[-\left(\frac{v}{1-\rho^2}\right)\right] \left(\exp\left[\left(\frac{\rho\sqrt{v^2-w^2}}{1-\rho^2}\right)\right] + \exp\left[-\left(\frac{\rho\sqrt{v^2-w^2}}{1-\rho^2}\right)\right]\right)$$

11. Let $N \sim \text{Poisson}(\lambda)$. Conditional on $N = n$, define

$$X \mid (N = n) \sim \text{Binomial}(n, p).$$

Find the marginal distribution of X and show that

$$X \sim \text{Poisson}(p\lambda).$$

Solution: We are given that

$$P(X = k) = \sum_{n=k}^{\infty} P(X = k \mid N = n)\, P(N = n).$$

Since $X \mid (N = n) \sim \text{Binomial}(n, p)$ and $N \sim \text{Poisson}(\lambda)$, we have

$$P(X = k \mid N = n) = \binom{n}{k} p^k (1-p)^{n-k}, \quad P(N = n) = e^{-\lambda}\frac{\lambda^n}{n!}.$$

Therefore,

$$P(X = k) = \sum_{n=k}^{\infty} \binom{n}{k} p^k (1-p)^{n-k} e^{-\lambda}\frac{\lambda^n}{n!}.$$

Simplify the combinatorial term:

$$\binom{n}{k}\frac{\lambda^n}{n!} = \frac{\lambda^k}{k!} \cdot \frac{\lambda^{n-k}}{(n-k)!}.$$

Substituting this back,

$$P(X = k) = e^{-\lambda}\frac{(p\lambda)^k}{k!} \sum_{n=k}^{\infty} \frac{[(1-p)\lambda]^{n-k}}{(n-k)!}.$$

Let $m = n - k$. Then

$$P(X = k) = e^{-\lambda}\frac{(p\lambda)^k}{k!} \sum_{m=0}^{\infty} \frac{[(1-p)\lambda]^m}{m!}.$$

The summation is the exponential series:

$$\sum_{m=0}^{\infty} \frac{[(1-p)\lambda]^m}{m!} = e^{(1-p)\lambda}.$$

Hence,

$$P(X = k) = e^{-\lambda} e^{(1-p)\lambda}\frac{(p\lambda)^k}{k!} = e^{-p\lambda}\frac{(p\lambda)^k}{k!}.$$

This is the pmf of a Poisson$(p\lambda)$ random variable.

$$\boxed{X \sim \text{Poisson}(p\lambda).}$$

12. Let (X, Y) have a bivariate normal distribution. Which of the following statements must be true? 1. Any nonzero linear combination of X and Y has a normal distribution. 2. $E(Y \mid X = x)$ is a linear function of x. 3. $\text{Var}(Y \mid X = x) \leq \text{Var}(Y)$.

Solution: Let (X, Y) be jointly normal with means μ_X, μ_Y, variances σ_X^2, σ_Y^2, and correlation ρ.

(I) For any constants a, b not both zero, the linear combination $aX + bY$ is normally distributed, since every linear combination of jointly normal random variables is normal. Hence, statement (I) is true.

(II) For a bivariate normal distribution, the conditional distribution of Y given $X = x$ is

$$Y \mid X = x \sim N\left(\mu_Y + \frac{\text{Cov}(X, Y)}{\sigma_X^2}(x - \mu_X),\ \sigma_Y^2(1 - \rho^2)\right).$$

Thus,

$$E(Y \mid X = x) = \mu_Y + \frac{\rho\sigma_Y}{\sigma_X}(x - \mu_X),$$

which is a linear function of x. Hence, statement (II) is true.

(III) From the same conditional formula,

$$\text{Var}(Y \mid X = x) = \sigma_Y^2(1 - \rho^2).$$

Since $0 \leq (1 - \rho^2) \leq 1$, we have

$$\text{Var}(Y \mid X = x) \leq \sigma_Y^2 = \text{Var}(Y),$$

with equality only when $\rho = 0$. Hence, statement (III) is also true.

Therefore, all three statements hold.

$$\boxed{\text{All of (I), (II), and (III) are true.}}$$

13. Let random variables Y_1, Y_2, Y_{12} be independent, where

$$Y_i \sim \text{Poisson}(\lambda_i), \quad i \in \{1, 2, 12\}.$$

Define

$$X_1 = Y_1 + Y_{12}, \qquad X_2 = Y_2 + Y_{12}.$$

(a) Find the covariance $\text{Cov}(X_1, X_2)$.

(b) Derive the joint probability mass function (pmf) of (X_1, X_2).

(c) Comment on the independence of X_1 and X_2 when $\lambda_{12} = 0$.

Solution:

(a) Covariance Using linearity of covariance and the independence of Y_1, Y_2, Y_{12},

$$\begin{aligned}\text{Cov}(X_1, X_2) &= \text{Cov}(Y_1 + Y_{12},\ Y_2 + Y_{12}) \\ &= \text{Cov}(Y_{12}, Y_{12}) \\ &= \text{Var}(Y_{12}) = \lambda_{12}.\end{aligned}$$

Hence, the covariance between X_1 and X_2 is determined solely by the shared Poisson component Y_{12}.

(b) Joint pmf For $x_1, x_2 \in \{0, 1, 2, \dots\}$, we use the law of total probability conditioning on $Y_{12} = k$:

$$\begin{aligned} P(X_1 = x_1, X_2 = x_2) &= \sum_{k=0}^{\min(x_1,x_2)} P(Y_{12} = k)\, P(Y_1 = x_1 - k)\, P(Y_2 = x_2 - k) \\ &= \sum_{k=0}^{\min(x_1,x_2)} \left(e^{-\lambda_{12}} \frac{\lambda_{12}^k}{k!}\right) \left(e^{-\lambda_1} \frac{\lambda_1^{x_1-k}}{(x_1-k)!}\right) \left(e^{-\lambda_2} \frac{\lambda_2^{x_2-k}}{(x_2-k)!}\right) \\ &= e^{-(\lambda_1+\lambda_2+\lambda_{12})} \sum_{k=0}^{\min(x_1,x_2)} \frac{\lambda_{12}^k\, \lambda_1^{x_1-k}\, \lambda_2^{x_2-k}}{k!\,(x_1-k)!\,(x_2-k)!}. \end{aligned}$$

Thus, the joint pmf of (X_1, X_2) is

$$\boxed{P(X_1 = x_1, X_2 = x_2) = e^{-(\lambda_1+\lambda_2+\lambda_{12})} \sum_{k=0}^{\min(x_1,x_2)} \frac{\lambda_{12}^k\, \lambda_1^{x_1-k}\, \lambda_2^{x_2-k}}{k!\,(x_1-k)!\,(x_2-k)!}.}$$

(c) Special case $\lambda_{12} = 0$ If the shared component has mean zero, then $Y_{12} \equiv 0$ with probability 1. Hence,

$$X_1 = Y_1, \qquad X_2 = Y_2.$$

Because Y_1 and Y_2 are independent,

$$P(X_1 = x_1, X_2 = x_2) = \left(e^{-\lambda_1} \frac{\lambda_1^{x_1}}{x_1!}\right) \left(e^{-\lambda_2} \frac{\lambda_2^{x_2}}{x_2!}\right),$$

and therefore $\mathrm{Cov}(X_1, X_2) = 0$. This confirms that X_1 and X_2 are independent when $\lambda_{12} = 0$.

$$\boxed{\mathrm{Cov}(X_1, X_2) = \lambda_{12}, \qquad (X_1, X_2) \text{ jointly Poisson as above.}}$$